BH Thermodynamics Programme: Overview of the Six-Paper Research Programme

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Claim 1: Asymptotic constant $c = 8/(3\pi)$ for unit-cube packing in a 4D ball. For a four-dimensional ball of radius $R = 2k + 1$, the count $N(k)$ of integer-centred unit cubes contained in it gives a volume deficit $\Delta(R) = V_4(R) - N(k)$ whose leading constant is derived in closed form by inclusion–exclusion: $c = 8/(3\pi) \approx 0.84883$. Numerical verification agrees to within 0.024%. (Paper 3)

Claim 2: Exact agreement with 4+1-dimensional Tangherlini black hole thermodynamics. Self-consistent determination of the central-projection metric yields $R \propto M^{1/2}$, Hawking temperature $T_H \propto M^{-1/2}$, lifetime $\tau \propto M^{3/2}$, and evaporation rate $dM/dt \propto -M^{-1/2}$. The ratio with the Bekenstein–Hawking entropy is $\Delta(R)/S_{BH} \to 32/(3\pi) \approx 3.40$. (Papers 4–5)

Claim 3: Independent path without holographic assumptions. The programme does **not** assume the 't Hooft–Susskind holographic principle, the Maldacena AdS/CFT correspondence, or any specific quantum-gravity programme (LQG/CDT/string theory). It derives the above results from minimal geometric and combinatorial assumptions only — central projection and Planck-scale discreteness. Zero self-citation. (Papers 1–6)

Paper 1: Overview and Hypotheses From premises A (entropy is geometric), B (evaporation is horizon-local), and C (spacetime may be discrete at the Planck scale), two conditional propositions are extracted: Hypothesis 1 (Projection) and Hypothesis 2 (Discrete Drift).

Paper 2: Mathematical Foundations of Central Projection Gnomonic projection from $S^4(R) \subset \mathbb{R}^5$ to a four-dimensional tangent hyperplane. The induced metric $g_{\mu\nu} = (R^2/\ell^2)(\delta_{\mu\nu} - x_\mu x_\nu/\ell^2)$ satisfies $G_{\mu\nu} + (3/R^2)g_{\mu\nu} = 0$, becomes de Sitter under Lorentzian continuation, and Schwarzschild–de Sitter with a mass parameter.

Paper 3: Mathematical Foundations of Discreteness Packing condition $\sum(|c_i| + 1/2)^2 \leq R^2$. Exact computation of $N(k)$ for $k = 0..60$ (with $N(60) = 1{,}028{,}515{,}513$). Asymptotic expansion $\Delta(R) = (16\pi/3)R^3 - 6\pi R^2 + O(R)$. Connection to the Lagrange–Jacobi four-square representation count $r_4(N) = 8\sigma(N)$.

Papers 4 and 5: Physical Connection and Dynamics Constant ratio $32/(3\pi)$ between $\Delta(R)$ and the 4+1 Bekenstein–Hawking entropy. From a Friedmann-like relation $E_{\text{total}} \propto R^2$, all Tangherlini scaling laws are recovered.

Paper 6: Synthesis and Open Problem The principal open problem — how the 4+1-dimensional results reduce to observed 3+1-dimensional black hole physics — is precisely formulated. Three candidate mechanisms: KK compactification, Randall–Sundrum/ADD brane localization, dS-CFT-style holographic projection.

DOIs of the Six Papers

Paper 1 (Overview)	10.5281/zenodo.19837588
Paper 2 (Projection)	10.5281/zenodo.19837590
Paper 3 (Packing)	10.5281/zenodo.19837592
Paper 4 (Entropy)	10.5281/zenodo.19837594
Paper 5 (Dynamics)	10.5281/zenodo.19837596
Paper 6 (Synthesis)	10.5281/zenodo.19837598

Correspondence with 4+1 Tangherlini

Quantity	Central-projection	4+1 Tangherlini
Mass–radius	$R \propto M^{1/2}$	$r_h \propto M^{1/2}$
Hawking temperature	$T_H \propto M^{-1/2}$	identical
Lifetime	$\tau \propto M^{3/2}$	identical
Entropy scaling	$\Delta(R) \propto R^3$	$S_{BH} \propto R^3$
Ratio	$\Delta/S_{BH} \to 32/(3\pi)$	

What This Research Does *Not* Claim

- That physical spacetime is 4+1-dimensional
- That higher-dimensional spacetime physically exists
- A first-principles derivation of the Bekenstein–Hawking entropy
- A replacement of holographic or quantum-gravity programmes

The "discrepancy" in the evaporation-rate exponent originally flagged in the programme outline is, in fact, a confusion between 3+1 and 4+1 conventions; in 4+1 dimensions the agreement is exact. The genuine open problem is the 4+1 → 3+1 reduction mechanism.

Note (English) Note (Japanese)

Zenn article https://zenn.dev/noriaki_kihara/articles/bh-thermodynamics-projection

Note (Japanese) https://note.com/kiharanoriaki/n/n4524006ef175

Note (English) https://note.com/kiharanoriaki/n/n916eac778c1c

Source code (GitHub) https://github.com/WurabeSeiji/ai-chat-logs-open/tree/main/BH

All papers are available on Zenodo as PDF, LaTeX, and Markdown (Japanese/English bilingual, CC BY 4.0).